# Spiral Matrix (medium)

## Problem Statement

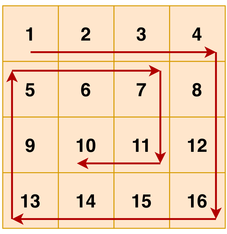
Given a 2D matrix of size `m x n`, return the 1D array containing all elements of matrix in `spiral order`.

### Examples

**Example 1**:

- **Input**: matrix =

```
[[1,2,3,4],
 [5,6,7,8],
 [9,10,11,12],
 [13,14,15,16]]
```




- **Expected Output**: `[1,2,3,4,8,12,16,15,14,13,9,5,6,7,11,10]`
- **Justification**: We have traversed the matrix in the spiral order.

**Example 2**:

- **Input**: matrix =

```
[[10,20,30],
 [40,50,60],
 [70,80,90]]
```

- **Expected Output**: `[10,20,30,60,90,80,70,40,50]`
- **Justification**: The traversal starts at the top-left, moves right to the end of the row, then down the right column, then left along the bottom row, up the left column, and finally captures the center.

**Example 3**:

- **Input**: matrix =

```
[[1,2],
 [3,4],
 [5,6]]
```

- **Expected Output**: `[1,2,4,6,5,3]`
- **Justification**: The traversal starts at the top-left, moves right, then down through the right column, and finally moves up the left column.

## Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.List;

public class Solution {
    public List<Integer> spiralOrder(int[][] matrix) {
        List<Integer> result = new ArrayList<>();
        // ToDo: Write Your Code Here.
        return result;
    }
}

```

Test Case | Run Code Result | Stdout

[[10,20,30],[40,50,60],[70,80,90]]

Use Example Testcases | Run | Submit